

ownership

which neuronal tree receives δ_u ?



correct tree

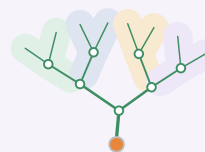
same bandwidth; different neuron-to-tree map



deranged tree

within-tree address

once the neuron is known, where should credit go?

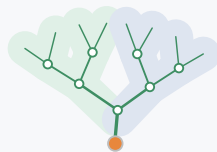


subtree routes refine $\delta_u \rightarrow \delta_{u,k}$



$K = 1$

one shared coordinate



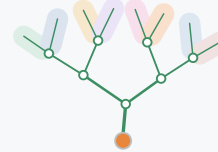
$K = 2$

two coarse subtrees



$K = 4$

intermediate routes



$K = 8$

full route resolution

—●— ancestry —■— deranged —▲— depth-interleaved —◆— random sparse —▶— random rank-K —✚— learned rank-K

physical stage count



same forward budget

multiplicative task hierarchy

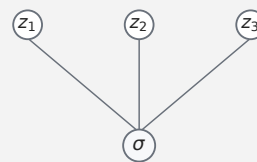
G_1 coarse factor

G_2 intermediate factor

G_3 fine factor

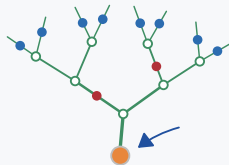
$y^* = s G_1 G_2 G_3$

representation-matched control



same routed fields

reconstructed arbor



mapped E/I contacts

ancestry route dictionary



nested subtrees define addresses

capacity per wire

dense —————

ancestry —————

control —————

model- and density-matched gains